

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
ASSESSMENT TOOL 1– Secure Infrastructure Architecture

Course Code:	CSA5802	Course Title:	DEVSECOPS ENGINEERING AND APPLICATION SECURITY
CO Assessed:	CO4 – Precision (BL1, BL2, BL3)	Assessment:	Assessment Tool 1
Weightage:	30% of CIA	Total Marks:	100
CO 4 Statement:	Design Secure Infrastructure Architecture using DiagrammeR or Drawio.		
Student Name:	Krithika M	Reg. No.:	192521385
Section / Batch:	B-Tech IT	Date:	19-08-2026

Instructions to Students

- Design a Secure Infrastructure Architecture for the selected application/system using DiagrammeR or Draw.io.
- Include essential components such as Firewall, WAF, Load Balancer, Web/Application Server, Database, IAM, IDS/IPS, and Monitoring/Logging wherever applicable.
- Clearly represent network zones, data flow, security controls, and trust boundaries in the architecture.
- Ensure the diagram is original, professionally formatted, clearly labelled, and submitted along with its editable source file.

Question Type	Focus Area	Questions	Marks
Secure Infrastructure Architecture	Conceptual Understanding	Q1	Q1 –100
GRAND TOTAL:		100 Marks	

Code of Conduct

I Krithika M / 1922521385(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Krithika M

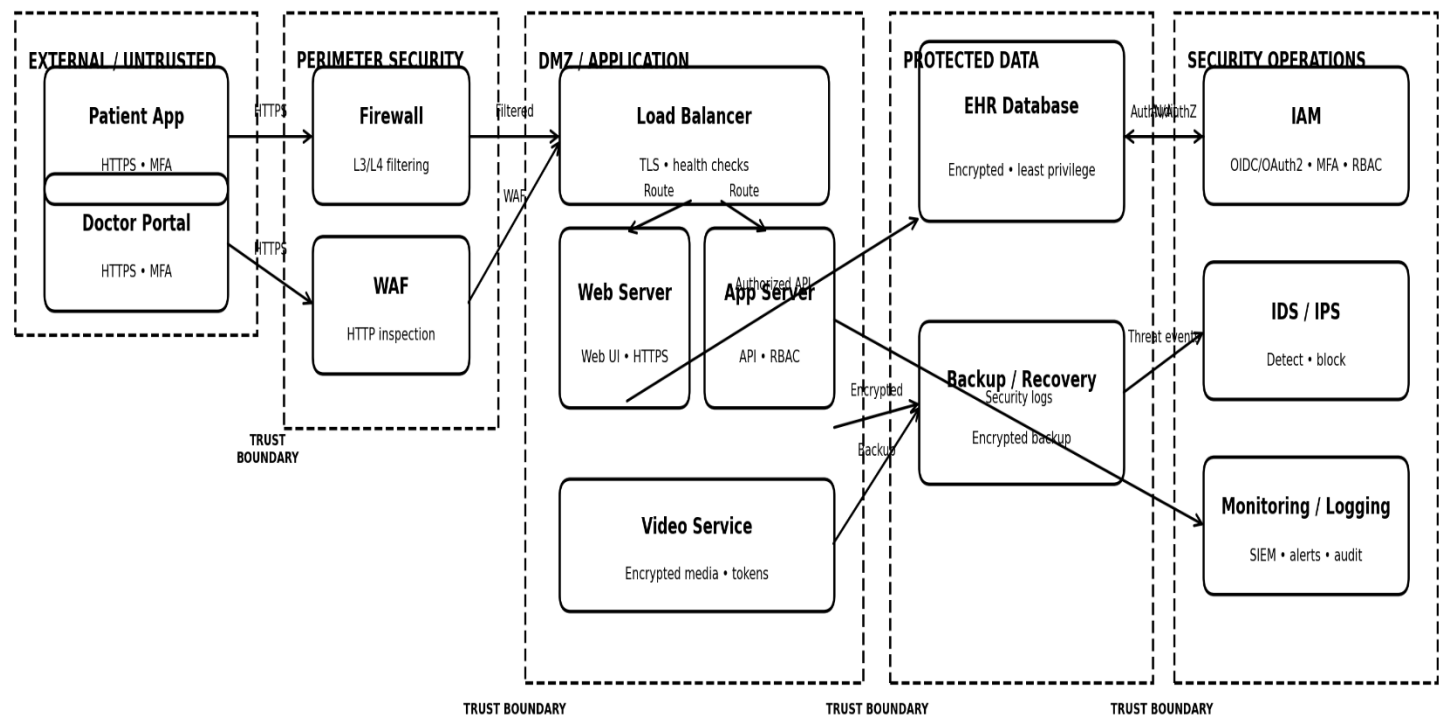
RUBRICS FOR CALCULATION

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Problem Context	20%	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	30%	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem-Solving Approach	20%	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	20%	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	10%	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand

Secure Infrastructure Architecture for a Healthcare Telemedicine Platform
Original Draw.io / DiagrammeR-style Architecture Design

SECURE HEALTHCARE TELEMEDICINE INFRASTRUCTURE

Network zones • data flow • security controls • trust boundaries



ENCRYPTION: TLS 1.3 in transit | AES-256 at rest | Managed key rotation

Design principle: no direct client-to-database access • least privilege • segmentation • continuous monitoring

Figure 1. Secure telemedicine infrastructure showing network zones, data flows, security controls, and trust boundaries.

1. Purpose

This architecture is designed for a healthcare telemedicine platform providing secure patient and doctor access to online consultations and electronic health records. It follows

defense-in-depth by separating external access, perimeter security, application services, protected data, and security operations.

2. Essential Components

Patient App / Doctor Portal: Secure client interfaces using HTTPS and MFA.

Firewall: Filters unauthorized network traffic at the perimeter.

WAF: Inspects HTTP/HTTPS traffic and protects web-facing services.

Load Balancer: Distributes approved traffic and performs health checks.

Web/Application Server: Hosts the web interface, APIs, business logic, and clinical workflows.

Video Service: Provides encrypted teleconsultation sessions with controlled tokens.

EHR Database: Stores healthcare records with encryption at rest and restricted access.

IAM: Provides authentication, MFA, OAuth2/OIDC, RBAC, and service identities.

IDS/IPS: Detects suspicious traffic and can block malicious activity.

Monitoring/Logging: Collects audit/security events and supports SIEM alerts and investigation.

Backup/Recovery: Maintains encrypted backups and supports restoration.

3. Network Zones

External / Untrusted Zone: Patient and doctor devices connect through the Internet using HTTPS.

Perimeter Security Zone: Firewall and WAF filter and inspect incoming traffic.

DMZ / Application Zone: Load balancer, web server, application server, and video service handle application functions.

Protected Data Zone: EHR database and backup systems are isolated from direct client access.

Security Operations Zone: IAM, IDS/IPS, monitoring, logging, and SIEM provide centralized security controls.

4. Data Flow and Trust Boundaries

Patient or doctor initiates a request through the appropriate application using HTTPS.

Traffic crosses the external-to-perimeter trust boundary and is filtered by the Firewall.

The WAF inspects web traffic and blocks or rejects suspicious requests.

Approved traffic reaches the Load Balancer, which routes requests to the Web/Application Server.

The Application Server validates identity and authorization through IAM before accessing protected resources.

The Application Server accesses the EHR Database through controlled service-to-service permissions; clients never connect directly to the database.

Video consultations use the dedicated Video Service with encrypted media and short-lived session tokens.

IDS/IPS monitors for suspicious activity while Monitoring/Logging collects audit and security events.

Security events are forwarded to centralized monitoring/SIEM for alerting, investigation, and compliance reporting.

5. Security Controls

Control	Implementation	Objective
Firewall	Network filtering and segmentation	Reduce unauthorized access
WAF	HTTP/HTTPS inspection	Protect web services
Load Balancer	Health checks and traffic distribution	Availability and controlled routing
IAM	MFA, OAuth2/OIDC, RBAC	Strong identity and least privilege
Encryption	TLS 1.3; AES-256 at rest	Confidentiality
IDS/IPS	Threat detection/prevention	Detect and block attacks
Monitoring/Logging	Central logs, SIEM, alerts	Detection and accountability
Database Security	Private zone and restricted accounts	Protect EHR information

6. Architecture Requirements Covered

Firewall, WAF, Load Balancer, Web/Application Server, Database, IAM, IDS/IPS, and

Monitoring/Logging are explicitly included.

Network zones are clearly separated and labelled.

Directional arrows show major application, data, authentication, and security-log flows.

Trust boundaries are marked between the major zones.

Sensitive EHR data is protected by encryption and least-privilege access.

The design prevents direct client-to-database communication.

7. Editable Source

The diagram is intentionally organized into editable-style boxes, connectors, zones, and labels so it can be recreated directly in Draw.io (diagrams.net) or DiagrammeR. The Word document contains the original architecture diagram and its complete specification.

For the final submission requirement of an editable source file, save/recreate the shown diagram in Draw.io as a .drawio file alongside this Word document.